Statement of Purpose

Zhuorui Ye, CS PhD Applicant

Research Experience and Vision. AI agents will be deployed in various sectors and interact with people, and with that come key challenges in ensuring their safe and human-beneficial deployment. As a result, my research interests broadly lie in machine learning, with the long-term vision of developing AI agents that are interpretable, robust, and helpful in real-world applications. To this end, I have worked on various related topics during my undergraduate research:

1. Make AI more interpretable to help humans better interact with AI. I have experience in developing methods for explainability and analyzing existing methods' properties from theoretical and empirical perspectives.

2. Build AI to tackle real-world challenges. I have worked on a few applications and developed novel solutions to address the real-world limitations of existing models.

Within these categories, I have also explored and feel excited about the possibilities brought by reinforcement learning (RL) plus foundation models. Foundation models provide rich world knowledge for general planning and decision-making, and RL plays an important role in allowing agents to improve from interactions continuously.

Make AI interpretable. I led a project on neural network interpretability in the ITCSC Summer Research Program at the Chinese University of Hong Kong, advised by Prof. Farzan Farnia. I analyzed the stability-fidelity properties of Smooth-Grad maps to help a principled application of the algorithm to different tasks. Our theoretical and empirical results give insights into whether to apply Gaussian smoothing on saliency maps based on the need for stability and fidelity. I also led the paper writing, resulting in a first-author paper that is under review at AISTATS 2025 [1]. This project improved my ability to rigorously analyze complex ML systems, which I enjoy leveraging even in my empirical work.

Reduce human annotation for RL interpretability algorithmically and with VLMs. Building upon this theoretical work, I was eager to explore interpretability in practice further. I co-led a project on concept-based models for RL at CMU, advised by Prof. Fei Fang and Prof. Geoff Gordon. In RL, concept-based models enable interpretability by mapping potentially high-dimensional inputs to actions through a bottleneck of human-understandable concepts. After an extensive literature review and initial experiments, we realized a crucial limitation of previous work is the assumption of unlimited access to a human expert for labels. To address this issue, we developed a training algorithm LICORICE that actively selects samples to query for human annotations, resulting in *thousands of times fewer labels than the state-of-the-art while maintaining performance*. We further explored using GPT-4o as the labeler in the pipeline and saw reasonable performance in a subset of environments, indicating the possibility of even no human efforts. The co-first author paper [2] was presented in four workshops including RLC InterpPol workshop (Oral) and ICML AutoRL workshop. The latest version has also been submitted to ICLR 2025. In addition to taking charge of experiments and leading project meetings, I iterated posters, Twitter posts, and slides to greatly hone my presentation skills, culminating in a well-received 50-minute talk at the Yao Class Seminar.

Throughout the extensive experiments, I have also identified my research style of employing rigorous empirical study and principled thinking when building a novel and useful system – I feel excited about continuously generating and experimentally validating new insights to eventually make things work, which represents more intellectual satisfaction than pure engineering. This experience also strengthened my commitment to developing AI that is both technically innovative and genuinely beneficial to human users.

Build AI for real-world challenges. My first research project was about 3D computer vision, resulting in a co-first author paper in AAAI 2024 [3]. I especially felt motivated when the professor depicted how our research is related to common applications like house robots and autonomous driving, and wanted to pursue work with practical impact.

Fascinated by AI's potential to solve meaningful real-world challenges, I have proposed the project of musical lyrics translation in collaboration with a fellow student. After communicating with a professional, we knew the existing song translation works often failed to generate fluent sentences. We also observed it was hard for GPT-4 to follow musical constraints in length and rhyme in a few-shot manner. These observations motivated us to focus on ensuring good translation quality while adhering to these constraints. We trained reward models to evaluate translation quality, proposed a two-stage training pipeline and inference-time optimization framework towards this goal. What started as a course project evolved through my initiative into a full research contribution, resulting in a co-first author paper [4] in EMNLP 2024 (Findings).

Build strategic language agents in real-world scenarios. To further explore the limits of current AI techniques in real-world scenarios, I have co-led a debate AI project aiming to achieve top-human-level competitive debate capabilities since this summer, advised by Prof. Fei Fang and Prof. Lei Li. Competitive debate requires the AI system to make strategic plans for persuading both its opponent and audience. Such persuasiveness is a core capability in complex decision-making in humans' daily lives. We mainly use ideas of game theory, for example, we build a minimax search tree to reason the potential interaction process and derive the most effective current claims. To ensure our design aligns with experts' knowledge, we iteratively update our AI agent based on human feedback from debate experts. Our current agent has outperformed the prompt-based baseline according to expert evaluation. As a future step, we envision performing self-play RL training to further optimize strategy selection during the rebuttal stage, including attacking the opponent's claims and reinforcing its own claims. This approach builds on my previous experience leading the RL training component in a related project with the same collaborators as the second author, where we enhanced LLMs' logical reasoning capabilities [5].

Community commitment. Having benefited greatly from mentors and inspiring talks throughout my journey, I am committed to giving back to the community through educational initiatives. To provide opportunities for students to communicate with and learn from each other, I have been hosting the weekly Yao Class seminar since the fall of 2024. As the host, I reach out to various speakers according to students' interests and am responsible for the logistics. I also teach competitive programming to high school students during holidays and have volunteered to share my academic and research experiences to guide sophomore and junior students in Yao Class.

Looking forward. Princeton's world-class research environment would be ideal for me to pursue a PhD journey and achieve my research goal. I admire the series of pioneering works of **Prof. Karthik R. Narasimhan** on AI agents, and hope to work with him to build reliable and helpful AI agents in the real world that learn from experiences and social interaction. I am also interested in working with **Prof. Danqi Chen** and **Prof. Sanjeev Arora** to advance researchers' understanding of current foundation models empirically and theoretically, exemplified by their recent work on understanding unintentional misalignment in DPO. Finally, I look forward to potentially collaborating with **Prof. Zhuang Liu**, as his research that gains rigorous empirical insights also aligns with my research style. Combining strong academic performance, competitive programming expertise, and demonstrated research capabilities, I am eager to contribute my unique research perspective and dedication to your department, looking forward to the potential collaborative achievements we can make.

References

[1] **Zhuorui Ye** and Farzan Farnia. On the trade-off between stability and fidelity of gaussian-smoothed saliency maps. *Submitted to AISTATS*, 2024.

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[3] Zifan Wang*, **Zhuorui Ye***, Haoran Wu*, Junyu Chen, and Li Yi. Semantic complete scene forecasting from a 4d dynamic point cloud sequence. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(6):5867–5875, Mar. 2024.

[4] **Zhuorui Ye***, Jinhan Li*, and Rongwu Xu. Sing it, narrate it: Quality musical lyrics translation. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 5498–5520, November 2024.

[5] Danqing Wang, **Zhuorui Ye**, Fei Fang, and Lei Li. Cooperative strategic planning enhances reasoning capabilities in large language models, 2024.